

SECTION C

INSTITUTE OF ENGINEERING & MANAGEMENT

Department of Computer Science & Engineering

MODULE 1 — PROJECT REPORT

AI-BASED AUTOMOTIVE REVIEW AND CUSTOMER
SENTIMENT ANALYTICS

AutoSense-NLP

A complete NLP-based system for automotive aspect extraction, transformer sentiment analysis, batch analytics, vehicle comparison, and interactive visualization.

Submitted by

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GitHub Repository: <https://github.com/Annwsha2024/AutoSense-NLP>

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Abstract

Automotive reviews contain detailed opinions about individual vehicle characteristics such as engine performance, comfort, mileage, safety, infotainment, price, battery range, charging, interior and exterior design. Conventional document-level sentiment analysis can reduce a review to one label even when the same review contains both positive and negative opinions. AutoSense-NLP addresses this problem by combining automotive-specific aspect extraction with transformer-based sentiment classification and clause-level association. The system detects 15 predefined automotive aspect categories, separates clauses around common contrast and conjunction terms, and applies the *cardiffnlp/twitter-roberta-base-sentiment-latest* model to classify sentiment as Positive, Negative, or Neutral. The model is used as a pretrained inference component rather than being presented as a model trained by this project.

The implementation is a Python application built with Pandas, PyTorch, Hugging Face Transformers, Scikit-learn, Pytest and Streamlit. It supports single-review analysis, batch CSV processing, analytics, and vehicle comparison. A 30-review synthetic dataset was used for controlled development and demonstration, while MuSe-CarASTE was used for dataset inspection and zero-shot sentiment evaluation. The final batch demonstration generated 67 aspect-level records: 48 Positive, 13 Negative and 6 Neutral. The sentiment component was evaluated on 14,574 labeled opinion annotations from the project copy of the MuSe-CarASTE annotations and achieved 49.98% accuracy and 0.5551 weighted F1. These results are interpreted as a domain-mismatch baseline rather than as evidence of a production-ready automotive classifier.

The project also includes 32 automated tests, all of which passed in the final verification run. The system demonstrates a complete, reproducible workflow from raw review text to aspect-level sentiment analytics while explicitly documenting the limitations of rule-based aspect extraction, heuristic clause association, and zero-shot use of a social-media-trained sentiment model.

Keywords: Aspect-Based Sentiment Analysis, Automotive Reviews, RoBERTa, Transformers, Streamlit, MuSe-CarASTE, NLP.

Source code: <https://github.com/Annweshha2024/AutoSense-NLP>

1. Introduction

Online automotive reviews provide a large source of qualitative information about vehicles. A single review may discuss several components and attributes in the same paragraph, making it difficult to extract actionable information manually. For example, the sentence “The battery range is excellent but charging takes too long” expresses two different opinions: a positive opinion about battery range and a negative opinion about charging time. A useful analytics system therefore needs to preserve the connection between an opinion and the automotive aspect it describes.

AutoSense-NLP was developed as a practical NLP pipeline for this requirement. Rather than attempting to reproduce a complete research-grade ASTE system, the project separates the problem into understandable components: text preprocessing, aspect detection, clause segmentation, transformer sentiment inference, aspect-sentiment integration, batch processing, visualization, and comparison. This modular structure also makes the system easier to test and explain during academic evaluation.

Project repository: <https://github.com/Annwesha2024/AutoSense-NLP> — the complete source code, tests, documentation and Streamlit application are available in this public GitHub repository.

2. Objectives

- Build an end-to-end Python application for automotive review sentiment analytics.
- Detect automotive-specific aspects using a controlled 15-category taxonomy.
- Use a genuine transformer sentiment model to classify Positive, Negative and Neutral sentiment.
- Associate sentiment with detected aspects and identify mixed sentiment within a review.
- Process multiple reviews from CSV files and generate structured aspect-level output.
- Provide an interactive Streamlit dashboard for analysis, analytics and vehicle comparison.
- Evaluate the pretrained sentiment component on labeled automotive opinion annotations.
- Validate the implementation with an automated test suite and document limitations clearly.

3. Background and Literature Review

Natural Language Processing (NLP) enables software to process and analyze human language. Sentiment analysis is a common NLP task in which text is categorized according to expressed polarity. In product-review applications, however, sentiment is often not uniform across the entire document. Aspect-Based Sentiment Analysis (ABSA) addresses this by considering the aspect being discussed, the associated opinion, and the sentiment expressed toward that aspect.

The MuSe-CarASTE dataset is particularly relevant because it was created for aspect sentiment triplet extraction in the automotive domain. The published work describes an automotive ASTE resource derived from automotive review video transcripts, with approximately 28,295 sentences organized into about 5,500 segments and annotations involving aspects, opinions and sentiment. The paper emphasizes that automotive reviews can contain complex aspect relationships and that the dataset is intended for ABSA/ASTE research [1].

For sentiment inference, this project uses CardiffNLP’s *twitter-roberta-base-sentiment-latest*. Its model card describes it as a RoBERTa-base model trained on approximately 124 million tweets and fine-tuned for sentiment analysis with TweetEval. The published labels are Negative, Neutral and Positive [2]. This domain is different from automotive reviews, which is an important limitation and is reflected in the evaluation results.

4. Dataset Description

Two types of data were used. First, a 30-row synthetic automotive review dataset was created specifically for controlled development, testing and demonstration. It contains `review_id`, `vehicle` and `review` fields and deliberately includes positive, negative, neutral, mixed and multi-aspect examples. It is not presented as real customer data.

Second, the project uses MuSe-CarASTE annotations for dataset inspection and evaluation. The local `FinalFullAnnotationsAllLabelsCompleted.csv` file contains 16,040 rows and 13 columns. Of these, 14,574 rows contain one of the valid sentiment labels — positive, negative or neutral — and 1,466 rows are unlabeled. The labeled portion contains 9,999 Positive, 2,578 Negative and 1,997 Neutral annotations. The project deliberately treats the ‘-’ label as unlabeled rather than incorrectly treating it as Neutral.

Table 1: Dataset files and observed statistics.

Dataset / File	Observed project statistics
Synthetic development CSV	30 reviews; 3 columns
MuSe train.csv	11,642 rows; 6 core columns

Dataset / File	Observed project statistics
MuSe devel.csv	4,398 rows; 6 core columns
FinalFullAnnotationsAllLabelsCompleted.csv	16,040 rows; 13 columns
Labeled annotation rows	14,574
Unlabeled rows	1,466
Unique raw aspects in labeled rows	2,853

The project uses the MuSe-CarASTE resource as an evaluation/reference source rather than claiming to reproduce its research models. The raw dataset files are excluded from the public GitHub repository, while attribution and acquisition instructions are retained in the project documentation.

5. Data Preprocessing and Aspect Taxonomy

5.1 Preprocessing

The preprocessing module validates the expected input columns, removes missing review text, collapses repeated whitespace, removes empty reviews and removes duplicate review text. The processed synthetic dataset is saved under data/processed. The design intentionally keeps preprocessing lightweight because excessive normalization can remove useful language cues needed by transformer models.

For the MuSe-CarASTE data, a separate dataset helper validates the six core fields: id, segment_id, review, aspect, opinion and sentiment. Only pos, neg and neu are considered valid sentiment labels. This prevents the unlabeled ‘-’ entries from contaminating evaluation statistics.

5.2 Automotive Aspect Taxonomy

The application maps automotive terms to 15 user-facing categories. The categories were chosen to cover common vehicle-review topics while remaining understandable in the dashboard.

Table 2: Automotive aspect categories and representative terms.

Category	Representative terms / examples
Vehicle	vehicle, car, automobile, SUV, sedan
Engine	engine, motor, cylinder, turbo
Battery	battery, battery pack, battery range
Mileage	mileage, fuel economy, kmpl, mpg
Safety	airbag, brake, ABS, collision, crash

Category	Representative terms / examples
Comfort	comfort, comfortable, seats, seat comfort
Service	service, maintenance, dealer, repair
Infotainment	infotainment, navigation, Bluetooth, screen
Price	price, cost, expensive, affordable
Design	design, styling, looks, appearance
Performance	performance, power, acceleration, handling
Reliability	reliable, durability, dependable, fault
Charging	charging, charger, charging time, fast charging
Interior	interior, dashboard, cabin, legroom, boot
Exterior	exterior, body, paint, wheels, headlights

5.3 Aspect Matching Strategy

Aspect detection is deliberately rule-based in the current implementation. Each category has a list of phrases, and matching uses regular expressions with word boundaries. This avoids false matches where a short keyword appears inside an unrelated word. The extractor can also return the exact matched phrases, which are displayed as matched text in the batch output.

The taxonomy was refined during testing to reduce obvious overlaps. For example, generic words such as “cheap” were not allowed to automatically imply Price because a review may say “cheap interior plastics.” Similarly, “responsive” was removed from the Performance vocabulary because it commonly describes infotainment systems. Seats were associated with Comfort to reduce unnecessary duplication with Interior.

6. System Design and Methodology

The system follows a modular pipeline in which each component has a clear responsibility. The design separates aspect detection from sentiment classification. This is important because sentiment and aspect identification are different tasks and can fail independently.

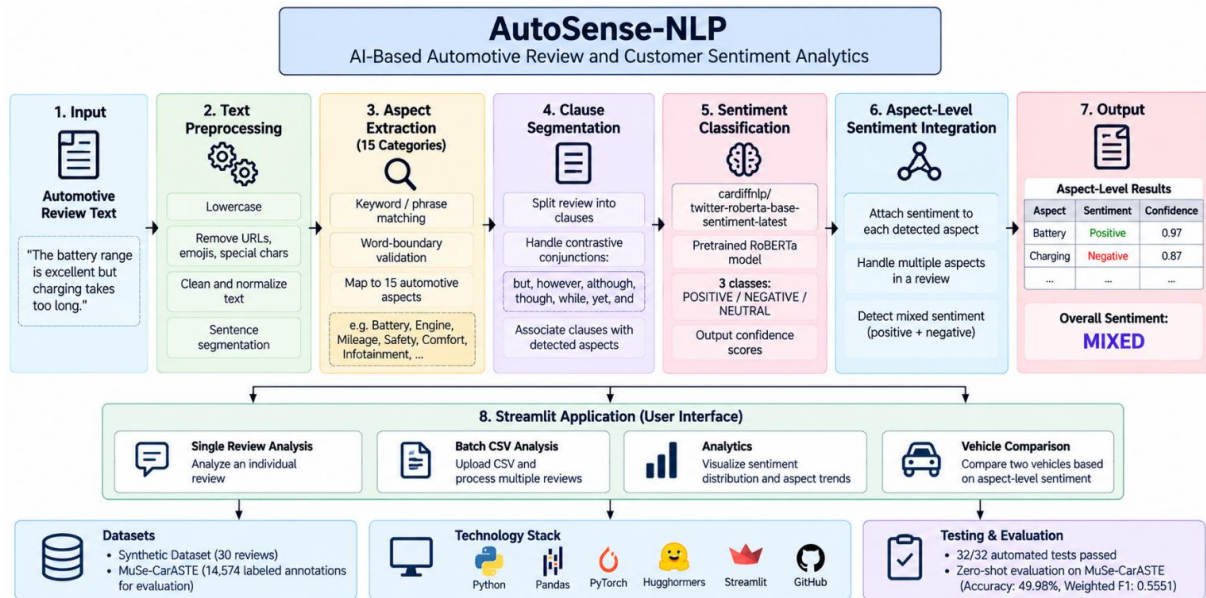


Figure 1: AutoSense-NLP processing architecture.

6.1 Review Processing Flow

- **Input:** A single review or a CSV containing automotive reviews.
- **Preprocessing:** Validate columns, remove missing/empty text, normalize whitespace and remove duplicate reviews.
- **Aspect detection:** Scan the review against the 15-category automotive vocabulary using word-boundary matching.
- **Clause segmentation:** Split text around conjunctions and contrast markers such as but, however, although, though, while, yet and and.
- **Sentiment inference:** Send each relevant clause to the RoBERTa classifier and obtain a class probability.
- **Aspect integration:** Attach the clause sentiment and confidence to each detected aspect.
- **Overall sentiment:** If both positive and negative aspect sentiments are present, the review is classified as Mixed; otherwise the dominant available polarity is used.
- **Output:** Structured records containing review, vehicle, aspect, sentiment, confidence, matched text and overall sentiment.

6.2 Mixed Sentiment Example

Consider the review “The battery range is excellent but charging takes too long.” The contrast marker “but” splits it into two clauses, each linked to its own aspect:

Table 3: Clause-level processing of a mixed-sentiment review.

Clause	Aspect	Sentiment	Confidence	Overall
The battery range is excellent	Battery	POSITIVE	0.9662	MIXED
charging takes too long.	Charging	NEGATIVE	0.8727	MIXED

The clause-level strategy is intentionally simple and interpretable. It is effective for clear constructions but cannot guarantee correct aspect-to-sentiment association for complex grammar, long-distance references, negation, sarcasm or sentences in which one opinion applies to multiple aspects.

7. Implementation

The implementation is organized as a small Python package plus a Streamlit application. The repository contains dedicated modules for preprocessing, dataset handling, aspect extraction, sentiment inference, aspect-sentiment integration, batch analysis and evaluation. Pytest tests are maintained separately. The complete source code is available on GitHub at <https://github.com/Annwesha2024/AutoSense-NLP>.

7.1 Transformer Sentiment Model

The sentiment component loads the tokenizer and sequence-classification model using Hugging Face Transformers and executes inference with PyTorch. The model is placed on the CPU in the current implementation, making the application runnable without a dedicated GPU. Batch inference is implemented by tokenizing multiple texts together, which reduces repeated model-call overhead during evaluation.

Table 4: Sentiment model configuration.

Property	Implementation
Model	cardiffnlp/twitter-roberta-base-sentiment-latest
Architecture	RoBERTa-base sequence classifier
Classes	Negative, Neutral, Positive
Inference	PyTorch + Hugging Face Transformers
Device	CPU
Maximum token length	512
Batch support	Yes

The model is not fine-tuned on the automotive dataset in this project. This distinction is important: the application uses a pretrained sentiment model as a reusable NLP component, while the automotive aspect layer is implemented separately. The model card identifies the model as English and Twitter/TweetEval-oriented, so transfer to automotive reviews is a domain adaptation problem rather than a guaranteed direct match [2].

7.2 Repository and Project Structure

The project is version-controlled and published at <https://github.com/Annwesh2024/AutoSense-NLP>. Figure 2 shows the repository's folder layout: raw and processed data under `data/`, reusable code under `src/`, automated tests under `tests/`, and configuration files (`.gitignore`, `pytest.ini`, `requirements.txt`, `README.md`) at the root.



Figure 2: Project folder structure (data, notebooks, outputs, src, tests and configuration files).

8. Application and Dashboard

A Streamlit interface was developed to make the NLP pipeline accessible without requiring users to run individual Python modules. The dashboard is organized into four modules, selectable from the Navigation sidebar: Single Review Analysis, Batch CSV Analysis, Analytics, and Vehicle Comparison. The screenshots in this chapter were captured from the running application.

8.1 Single Review Analysis

The user enters a review and triggers analysis. The interface displays the overall sentiment together with aspect-level rows containing the detected aspect, sentiment, confidence and relevant text. This view is intended for interactive demonstration and quick inspection of individual examples.

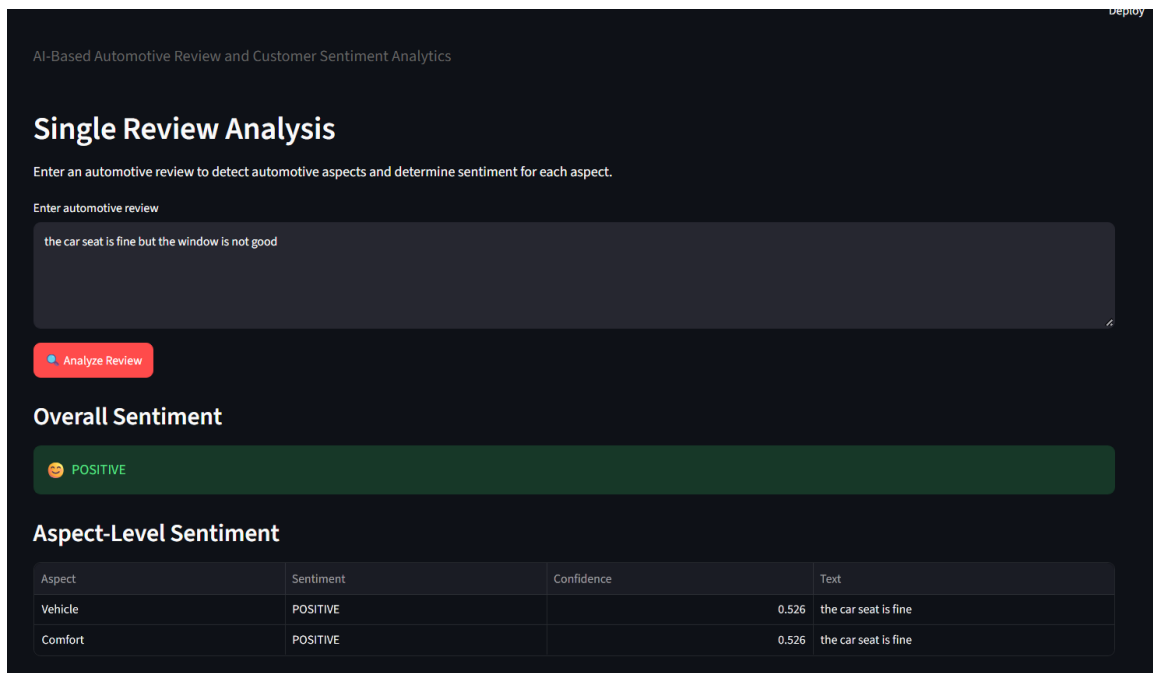


Figure 3: Single Review Analysis module showing overall sentiment and aspect-level results.

In the example above, the review “the car seat is fine but the window is not good” was analysed. The system detected the phrase “the car seat is fine” and assigned it to the Vehicle and Comfort aspects, both POSITIVE with confidence 0.526. The second clause about the window produced no aspect row because “window” is not part of the current 15-category taxonomy. This is a genuine illustration of the limitation of vocabulary-based aspect extraction discussed in Chapter 11, and it motivates the future scope of using a trained ABSA model.

8.2 Batch CSV Analysis

The batch interface accepts a CSV containing a review column and processes the records sequentially. A progress indicator provides feedback while the transformer is running. The resulting DataFrame can be viewed directly in the dashboard and downloaded as a CSV file.

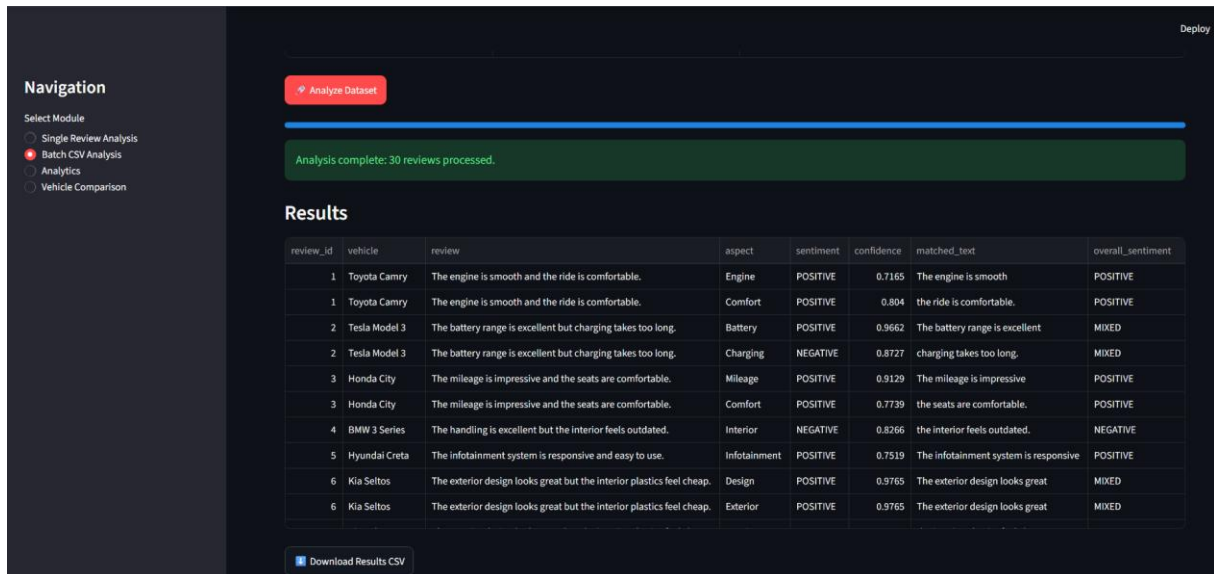


Figure 4: Batch CSV Analysis: 30 reviews processed, with aspect, sentiment, confidence, matched text and overall sentiment, and a Download Results CSV button.

As shown in Figure 4, a review containing both positive and negative aspects (for example, review 2 for the Tesla Model 3: Battery POSITIVE, Charging NEGATIVE) receives the overall label MIXED, whereas reviews with a single polarity keep that polarity as their overall sentiment.

8.3 Analytics

The analytics module summarizes the generated result file. It presents the number of reviews, number of aspect-level records, sentiment distribution, aspect distribution and average confidence. These views help identify which automotive topics are most frequently detected in the current dataset.

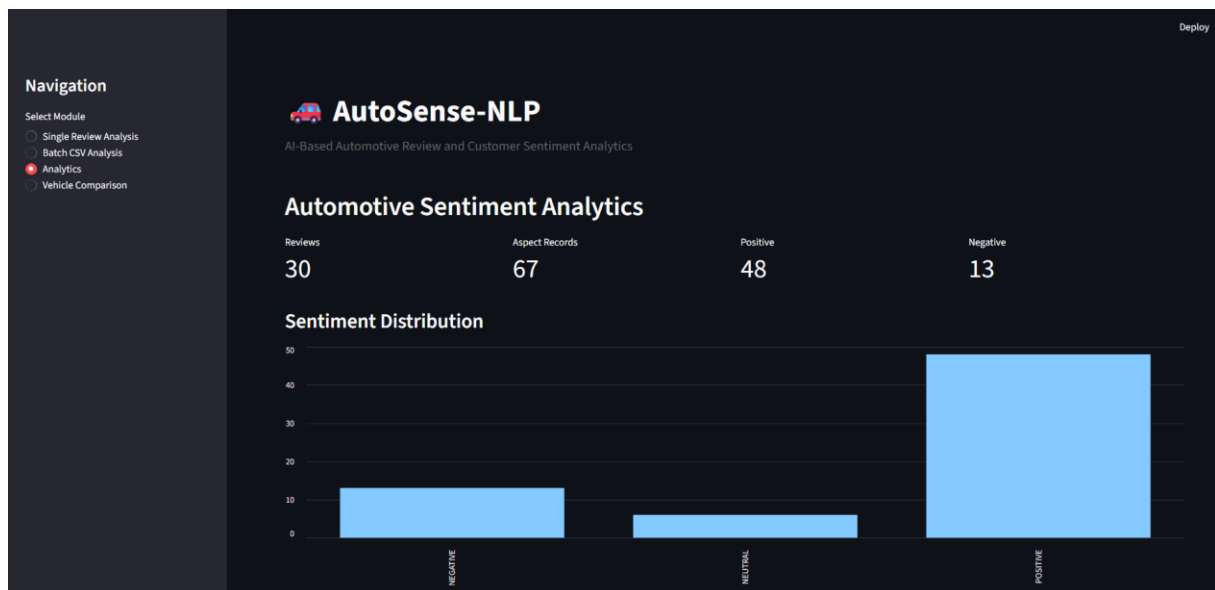


Figure 5: Analytics dashboard: 30 reviews, 67 aspect records (48 Positive, 13 Negative) and the sentiment distribution chart.



Figure 6: Analytics dashboard: aspect distribution across the 15 categories and average confidence by sentiment class.

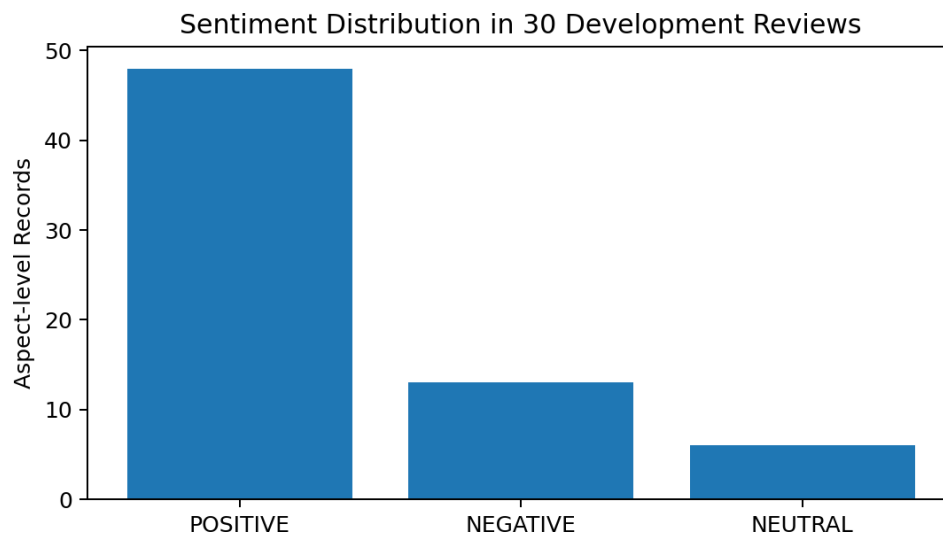


Figure 7: Final sentiment distribution for the 30-review development batch.

8.4 Vehicle Comparison

The comparison module allows two vehicles to be selected from the analyzed CSV. It reports review counts, aspect-record counts, sentiment distribution, aspect coverage, dominant sentiment by aspect and average confidence. The interface presents evidence rather than a hard-coded overall vehicle ranking; the dashboard itself states that the comparison “does not determine an overall winner between vehicles.”

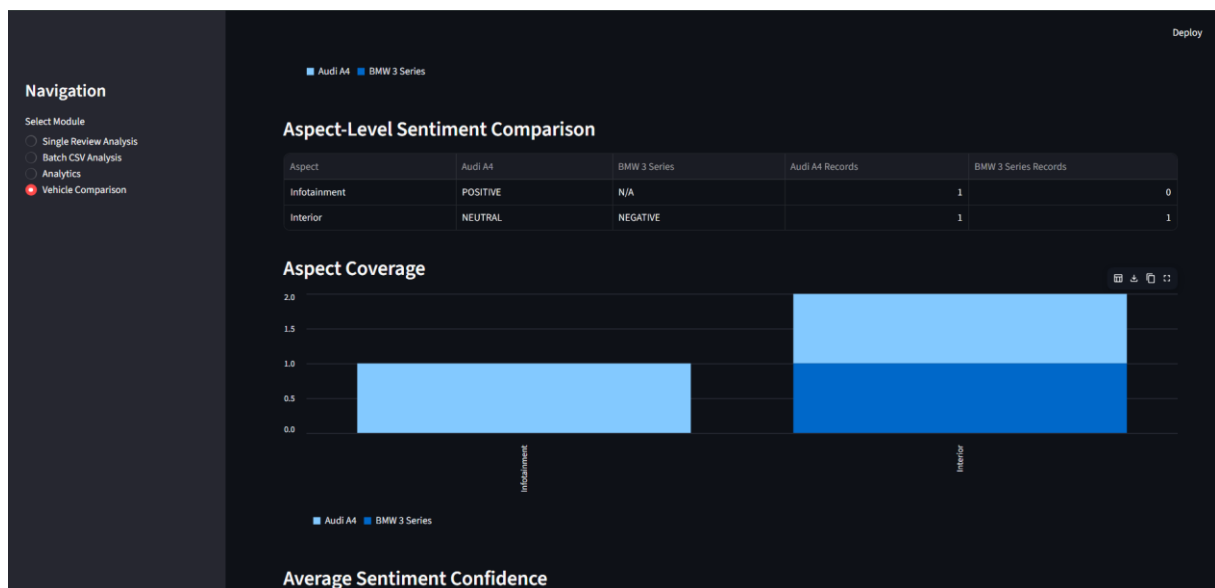


Figure 8: Vehicle Comparison (Audi A4 vs BMW 3 Series): aspect-level sentiment table and aspect coverage chart.

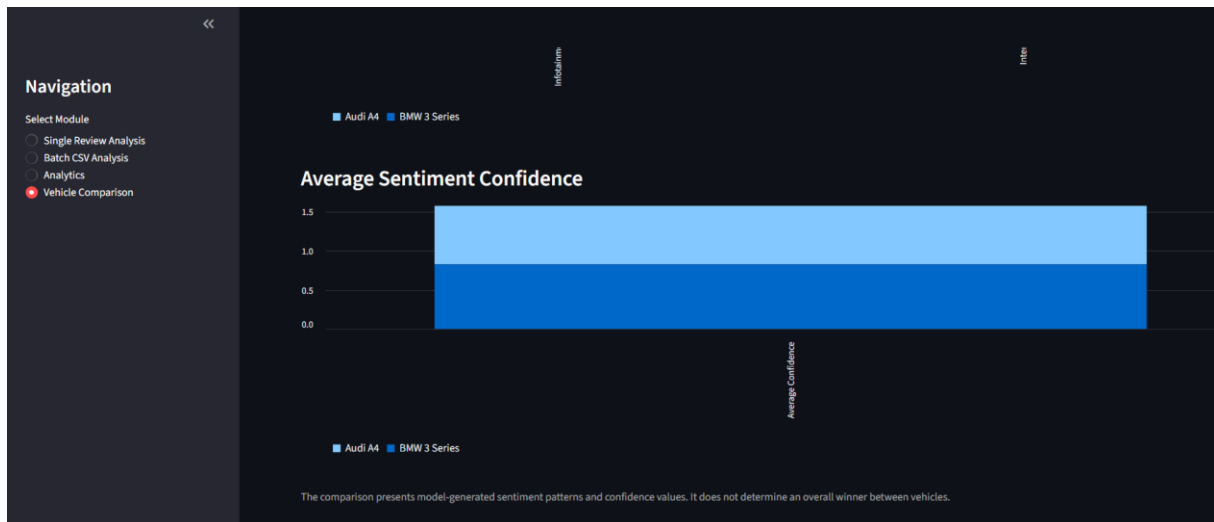


Figure 9: Vehicle Comparison: average sentiment confidence for the two selected vehicles.

The synthetic dataset is intentionally small, so vehicle-level counts should be interpreted as a demonstration of system functionality rather than a statistically representative market comparison.

9. Testing and Results

9.1 Automated Testing

Testing was performed at both component and system levels. The final repository contains 32 automated tests covering aspect extraction, aspect-sentiment integration, dataset validation, preprocessing and sentiment-model behavior. The final run completed with all 32 tests passing in 16.21 seconds.

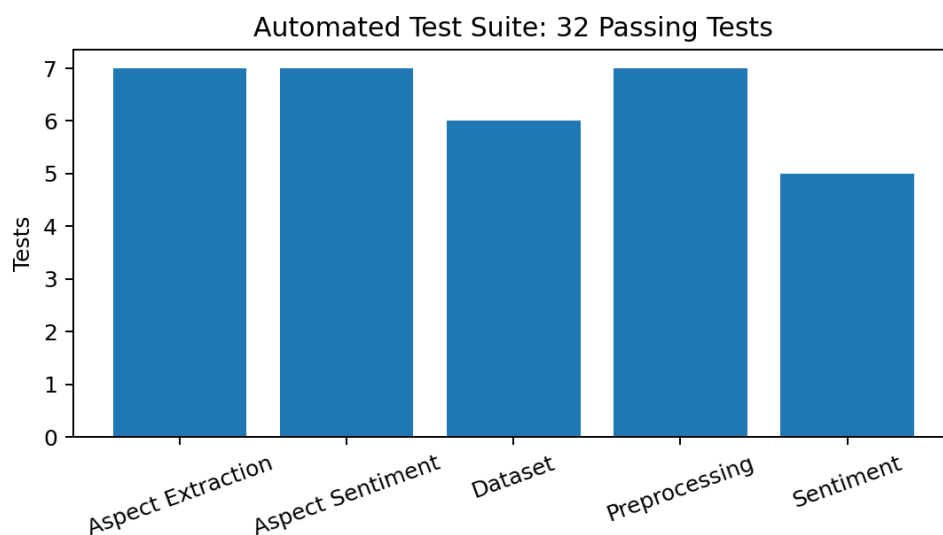


Figure 10: Breakdown of the 32 automated tests.

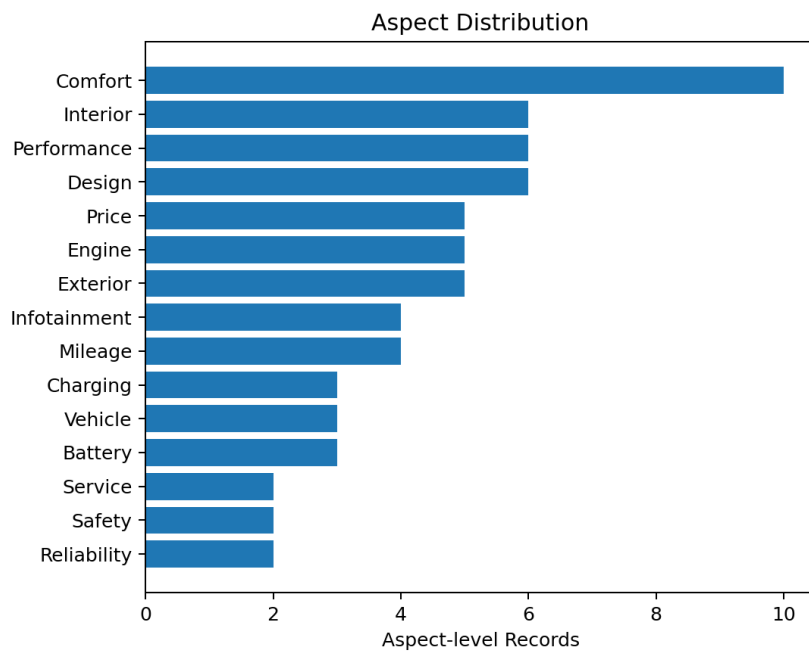
Table 5: Automated test results by area.

Test Area	Tests	Final status
Aspect Extraction	7	7/7 passed
Aspect Sentiment	7	7/7 passed
Dataset	6	6/6 passed
Preprocessing	7	7/7 passed
Sentiment	5	5/5 passed
Total	32	32/32 passed

9.2 Final Batch Results

Table 6: Results of the 30-review batch run.

Measure	Result
Input reviews	30
Aspect-level records	67
Positive records	48
Negative records	13
Neutral records	6

**Figure 11:** Aspect distribution in the final 30-review batch.

Comfort was the most frequently detected category with 10 records. Interior, Design and Performance each produced 6 records, followed by Engine, Exterior and Price with 5 records each. Battery and Charging each produced 3 records in the synthetic development batch.

10. Model Evaluation and Discussion

The sentiment component was evaluated using the labeled portion of the project copy of FinalFullAnnotationsAllLabelsCompleted.csv. The evaluation uses the gold opinion text as the model input and therefore measures the sentiment classifier independently of the rule-based aspect extraction stage. It is a zero-shot evaluation because the pretrained sentiment model was not fine-tuned on these automotive annotations.

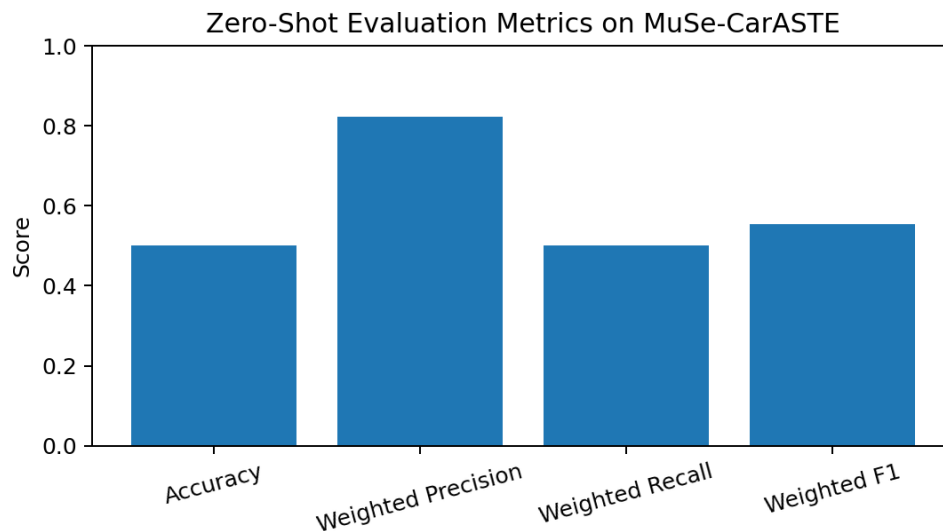


Figure 12: Zero-shot evaluation metrics on 14,574 labeled automotive opinion annotations.

Table 7: Overall (weighted) evaluation metrics.

Metric	Score
Accuracy	0.4998 (49.98%)
Weighted Precision	0.8219 (82.19%)
Weighted Recall	0.4998 (49.98%)
Weighted F1	0.5551 (55.51%)

Table 8: Per-class evaluation metrics.

Class	Precision	Recall	F1	Support
NEGATIVE	0.72	0.31	0.43	2,578
NEUTRAL	0.21	0.92	0.34	1,997
POSITIVE	0.97	0.47	0.63	9,999

10.1 Interpretation

The results show a strong precision value for Positive predictions but substantially lower recall for Positive and Negative classes, while Neutral recall is high. The weighted F1 of 0.5551 indicates that the pretrained model provides a useful baseline but does not fully adapt to the

automotive-review domain. The domain mismatch is expected because the model was developed for Twitter/TweetEval sentiment rather than automotive customer reviews [2].

11. Limitations

- Aspect extraction is rule-based and cannot reliably discover unseen automotive aspects. In the Single Review demonstration, the “window” clause was missed because that term is outside the 15-category vocabulary.
- Some words, such as “range,” can have more than one interpretation and require contextual disambiguation.
- Clause splitting is heuristic and may fail on complex sentence structures.
- The sentiment model is pretrained on social-media text and was not fine-tuned on automotive reviews.
- The reported evaluation is sentiment-only and uses gold opinion text; it is not an end-to-end ABSA score.
- The 30-review synthetic batch is intended for development and demonstration, not market-level statistical conclusions.

12. Conclusion and Future Scope

12.1 Conclusion

AutoSense-NLP successfully implements a complete automotive review sentiment analytics pipeline. The project combines lightweight rule-based aspect detection with a genuine transformer sentiment model and integrates the two into a practical application. It supports individual review analysis, batch processing, structured CSV output, analytics and vehicle comparison through a Streamlit dashboard.

The final implementation was validated with 32 automated tests, all of which passed. On the 30-review development dataset, the system produced 67 aspect-level records consisting of 48 Positive, 13 Negative and 6 Neutral predictions. The separate zero-shot evaluation on 14,574 labeled MuSe-CarASTE opinion annotations produced 49.98% accuracy and 0.5551 weighted F1. These numbers are reported transparently and should be interpreted as a baseline for a pretrained general-domain sentiment model rather than as a claim of production-grade automotive sentiment performance.

12.2 Future Scope

- Fine-tune a transformer on automotive-domain sentiment annotations.

- Replace keyword-based aspect extraction with a trained ABSA or ASTE model.
- Introduce contextual disambiguation for terms such as range, power, service and design.
- Improve clause-level sentiment association using dependency parsing or span-level models.
- Add multilingual support for Indian-language automotive reviews.
- Add temporal analytics to identify changes in customer sentiment over time.
- Integrate larger real-world review datasets and independent test sets.
- Add explainability features that highlight words or spans influencing sentiment predictions.
- Deploy the dashboard as a cloud application with secure dataset ingestion.
- Benchmark multiple transformer models and compare domain adaptation strategies.

References

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Appendix

Representative input: “The battery range is excellent but charging takes too long.”

Representative output: Battery → POSITIVE; Charging → NEGATIVE; Overall → MIXED. This example demonstrates why aspect-level sentiment is more informative than assigning a single sentiment label to the whole review.

Repository: AutoSense-NLP — <https://github.com/Annweshha2024/AutoSense-NLP>

Final verification: Git branch main synchronized with the GitHub repository; working tree clean; 32/32 tests passed.